RHINE-RUHR AREA

Oncology Referral Networks

Data-Driven Inflow Optimization & Board Efficiency at RKAM Hospital



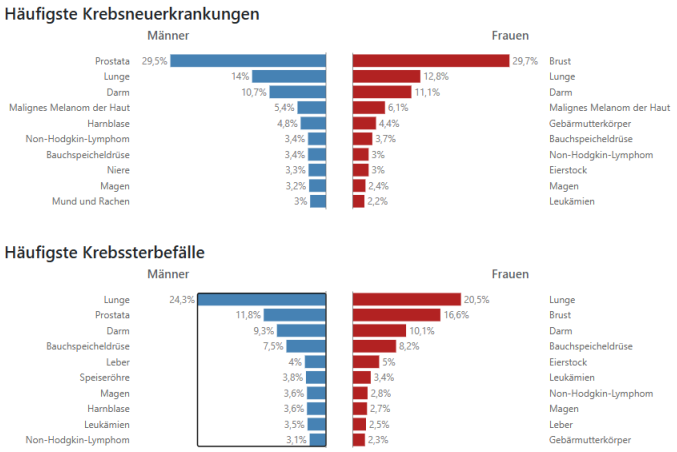
Final Project

Healthcare Data Engineering & Business Intelligence

Presenter: Birgit Drews

Clinical & Administrative Bottlenecks





In-Person Overload

Manual paper referrals slowing down patient intake.

Missing Documentation

Incomplete medical records causing critical diagnostic delays.

Capacity Overload

Clinical beds and Standard Tumor Board slots blocked by less severe Stage I & II cases.

Consequence

High-risk Stage IV oncology patients facing unacceptable waiting times.

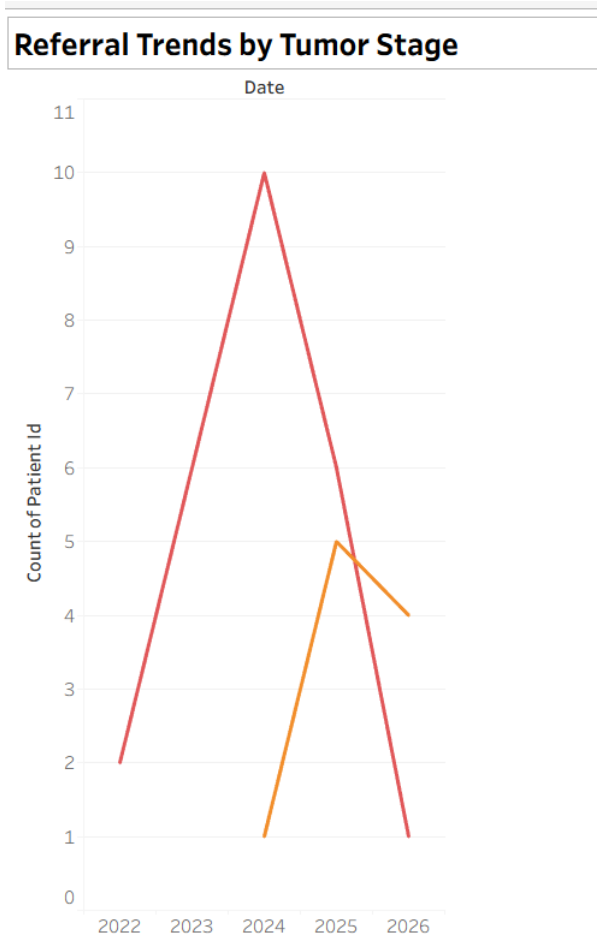
Schematic overview. Cancer centre in Essen.

Standard vs. Molecular Tumor Board

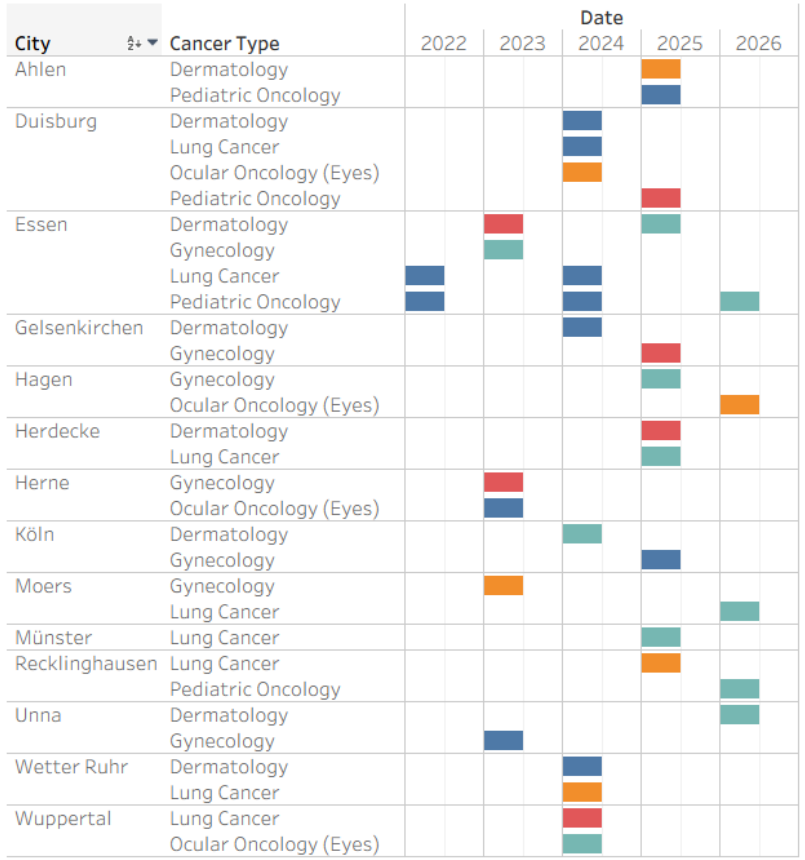
Referrals by UICC stage

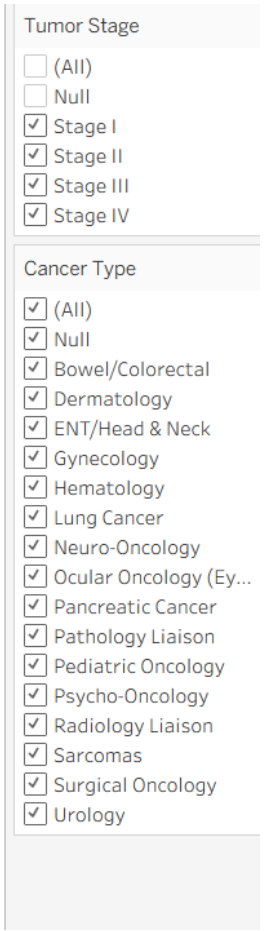
Board Typ	Zuweiser Typ	UICC Stadium			
		Stadium I	Stadium II	Stadium III	Stadium IV
Molekulares Tumorboard (MTB)	Hospital (Regional)	22	25	70	70
	Klinik (National)	4	7	17	7
	Kooperation Forsch..	5	17	37	41
	Praxis (Regional)	24	30	80	76
	Praxis (Überregional)	3	4	4	10
	Zentrum Ambulanz	2	5	14	11
Standard Tumorboard	Hospital (Regional)	103	108	27	24
	Klinik (National)	29	33	10	9
	Kooperation Forsch..	56	68	6	12
	Praxis (Regional)	142	107	35	22
	Praxis (Überregional)	23	21	3	4
	Zentrum Ambulanz	19	29	4	3

Board comparison

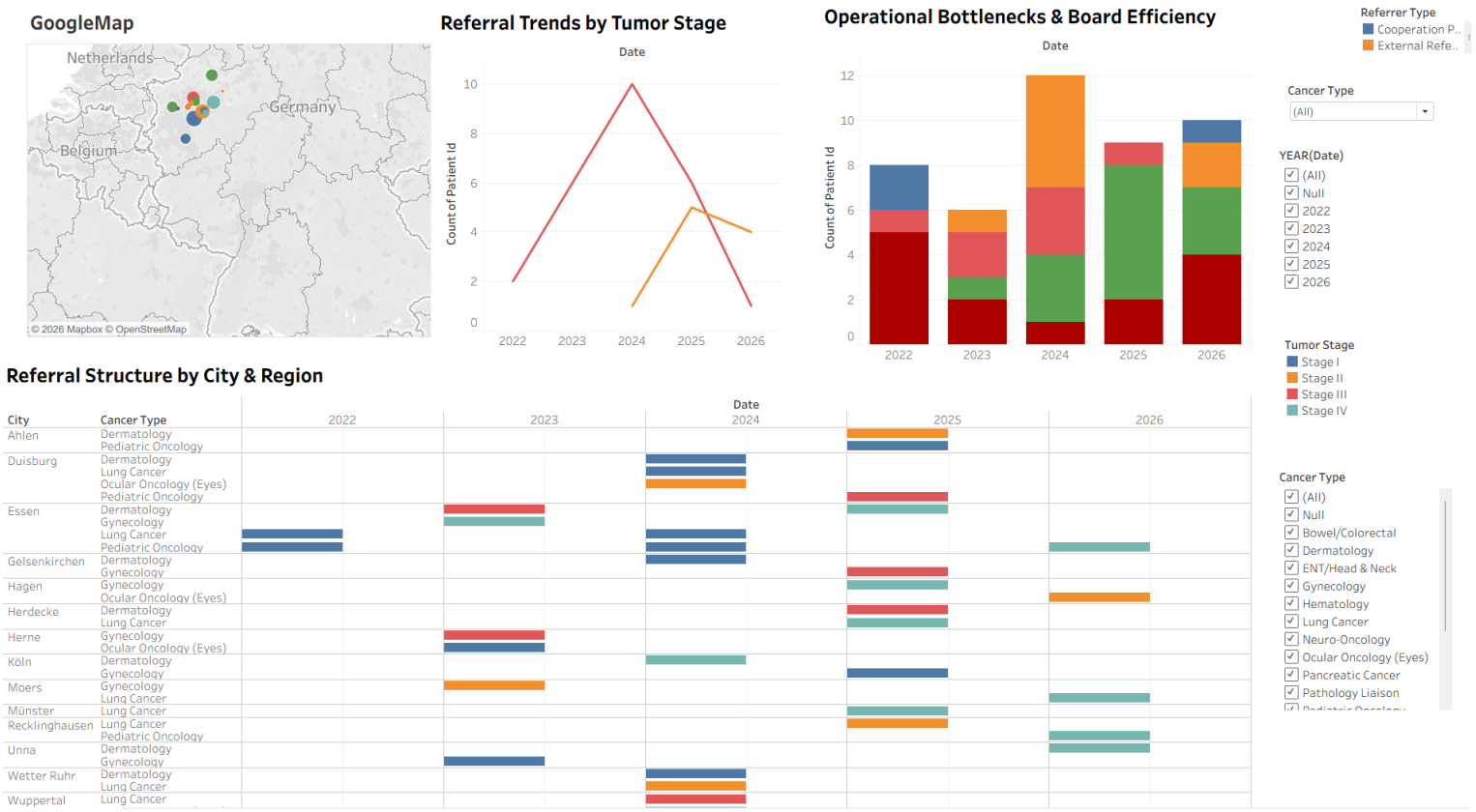


Referrer Development by Entity (2022-2026)





Live Interactive Dashboard & Healthcare Impact



Thank you

for your attention